PROFESSIONAL PROJECT PROPOSAL (Version 2.0)

Deep Learning-Based Rainfall Data Imputation and Prediction System

1.0 PROJECT OVERVIEW

This proposal outlines the development of a comprehensive deep learning solution for rainfall data management, including missing value imputation using Generative Adversarial Networks (GAN) and rainfall intensity prediction using Temporal Convolutional Networks (TCN).

2.0 PROJECT OBJECTIVES

Objective	Description	Technology
2.1 Data Imputation	Fill missing values in incomplete daily rainfall dataset using deep learning	Generative Adversarial Network (GAN)
2.2 Rainfall Prediction	Predict rainfall intensity at multiple temporal scales	Temporal Convolutional Network (TCN)

3.0 DATASET SPECIFICATIONS

Parameter	Details
Data Type	Daily rainfall measurements
Time Period	2009 - 2024 (16 years)
Number of Stations	3 telemetry stations
Dataset Dimensions	5,840 rows × 3 columns
Data Challenge	Incomplete dataset with missing values

4.0 METHODOLOGY

4.1 Phase 1: Data Preprocessing and GAN-Based Imputation

- Data cleaning and exploratory analysis

- Missing value pattern identification
- GAN model architecture design and training
- Validation of imputed values
- Quality assessment of completed dataset

4.2 Phase 2: TCN-Based Rainfall Prediction

- Feature engineering and sequence preparation
- TCN model development for multi-scale prediction
- Training and validation for daily, weekly, and monthly forecasts
- Model performance evaluation
- Hyperparameter optimization

5.0 DELIVERABLES

Deliverable	Description	Format
5.1 Complete Dataset	Rainfall data with imputed missing values	CSV/Excel
5.2 GAN Model	Trained model for data imputation	Python code + model files
5.3 TCN Model	Trained prediction model for rainfall intensity	Python code + model files
5.4 Prediction Results	Day, week, and month rainfall predictions	CSV/Excel + Visualizations
5.5 Technical Report	Comprehensive documentation of methodology and results	PDF document
5.6 Source Code	Complete well-documented Python code	Python scripts

6.0 PROJECT TIMELINE

Phase	Activity	Duration
Phase 1	Data preprocessing and GAN development	2 weeks
Phase 2	GAN training and validation	1 week
Phase 3	TCN model development	1 week
Phase 4	TCN training and multi-scale prediction	1 week
Phase 5	Testing, documentation, and delivery	1 week

Total Duration: 8 weeks (Maximum completion: End of August)

7.0 INVESTMENT QUOTATION

Item	Description	Amount (MYR)
7.1	GAN-based data imputation system development	1,200
7.2	TCN-based rainfall prediction model development	1,200
7.3	Data preprocessing and feature engineering	500
7.4	Model validation and performance optimization	500
7.5	Documentation and technical report	400
	TOTAL PROJECT COST	**3,800**

Note: This quotation is structured at a highly competitive freelance rate to accommodate research or academic budget constraints while ensuring PhD-level deep learning implementation. Pricing can be adjusted based on specific requirements and scope modifications.

8.0 TERMS AND CONDITIONS

- **Payment Terms:**
  - **Commencement Fee (30%):** RM 1,140 payable upon project kickoff and data handover.
  - **Phase 1 Milestone (30%):** RM 1,140 payable upon successful completion and demonstration of the GAN-based data imputation.
  - **Phase 2 Milestone (30%):** RM 1,140 payable upon completion of the TCN prediction models and initial results review.
  - **Final Delivery (10%):** RM 380 payable upon handover of all source code, final documentation, and project sign-off.
- **Confidentiality:** All data and research findings will be kept strictly confidential.
- **Intellectual Property:** Models and code will be delivered to the client upon full payment.
- **Support:** 30 days post-delivery support for bug fixes and clarifications.

9.0 CONCLUSION

This project will provide a robust deep learning framework for handling incomplete rainfall data and generating accurate multi-temporal scale predictions. The proposed solution leverages state-of-the-art GAN and TCN architectures to ensure high-quality results suitable for PhD-level research.

We look forward to collaborating on this important research initiative.

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